AI Model Evaluation — Interview Notes

Goal: Concept clear karo aur interview mein 30–60 seconds mein confidently explain kar sako.

1. Knowledge & Reasoning ■

Simple meaning: Model ko facts pata hain aur woh multiple facts ko connect karke correct conclusion nikal sakta hai ya nahi.

2 main parts: Factual knowledge — model ko information pata hai? | Reasoning — multiple steps follow karke answer nikal sakta hai?

Example: A, B aur C facts diye → kya model logically D conclusion nikal sakta hai?

> **Interview answer:** “Knowledge and reasoning evaluates whether a model knows factual information and can combine multiple pieces of information to solve multi-step problems.”

2. Coding & Software Engineering ■

Simple meaning: Model sirf code generate karta hai ya actually working software bana/fix bhi kar sakta hai?

Important areas: Function generation • Bug fixing • Multi-file projects • Refactoring • CLI/terminal tasks • API/function calling

Example: GitHub repo + issue → root cause find → code modify → tests pass.

> **Interview answer:** “Coding evaluation checks whether an AI model can generate, debug, modify and maintain real-world software, not just produce code snippets.”

3. Mathematics ■

Simple meaning: Model accurate numerical + symbolic reasoning kar sakta hai ya nahi.

Levels: Basic arithmetic • Multi-step word problems • Competition-level problems • Advanced mathematics • Research-level reasoning

Important point: Math evaluation useful hai because answer usually correct ya incorrect hota hai — sirf convincing answer enough nahi.

> **Interview answer:** “Mathematical evaluation measures whether a model can perform accurate numerical and symbolic reasoning.”

4. Long Context ■

Simple meaning: Model bohat large input mein se relevant information effectively use kar sakta hai ya nahi.

Tests: Long document se fact find • Different parts ki information connect • Large document summarize/compare • Large codebase understand

Key concept: Context window ≠ Context understanding. 1M-token context support karna automatically yeh prove nahi karta ke model 1M tokens ki information effectively use bhi karega.

> **Interview answer:** “Long-context evaluation checks whether a model can retrieve, connect and reason over information distributed across very large inputs.”

5. Vision & Multimodal ■■

Simple meaning: Model text ke saath images/visual information ko understand aur reason kar sakta hai ya nahi.

Examples: Charts/graphs • Invoices • Scanned documents • Tables • Forms • Computer screens • Engineering diagrams

Example: Invoice ki image do → model seller, date aur total amount identify kare.

Interview answer: “Vision and multimodal evaluation checks whether a model can understand and reason over visual information together with text.”

6. Agentic & Tool Use ■■

Simple meaning: LLM sirf answer generate na kare, balki actions perform kare.

Agent workflow: User → Agent → Tool → Result → Reason → Next Tool → Final Answer

Agent can: Browse web • Call APIs • Execute code • Manipulate files • Use software • Perform multi-step tasks

Core loop: Plan → Select tool → Execute → Check result → Recover/adjust

Interview answer: “Agentic evaluation measures whether a model can use tools, plan multi-step tasks, execute actions, track progress and recover from failures.”

Most important concept: Knowledge ≠ Ability to Act. Traditional LLM: Question → Answer. Agent: Goal → Plan → Tools → Actions → Verification → Result.

7. Safety & Alignment ■■

Simple meaning: Model safe, truthful aur responsible behavior karta hai ya nahi.

Evaluation includes: Harmful requests • Jailbreaks • Prompt injection • Sycophancy • Truthfulness

Interview answer: “Safety and alignment evaluation checks whether a model behaves responsibly, resists adversarial attacks, avoids harmful behavior and remains truthful.”

8. Instruction Following ■

Simple meaning: User ne jo instructions di hain, model exactly unko follow karta hai ya nahi.

Tests: Word limit • Required format • JSON • Markdown • Specific headings • Tone/style • Ambiguous instructions

Example: User says “5 bullet points and valid JSON” → model ko dono constraints satisfy karne hain.

Interview answer: “Instruction following evaluates whether a model correctly follows explicit constraints such as format, length, structure, tone and content requirements.”

■ 8 Domains — One-Line Revision

Domain	Simple Meaning
Knowledge & Reasoning	Know + think
Coding	Write + fix real code
Mathematics	Accurate calculations/reasoning
Long Context	Understand huge inputs
Vision/Multimodal	Understand images + text
Agentic/Tool Use	Think + use tools + act
Safety/Alignment	Safe + truthful behavior
Instruction Following	Follow user's requirements

■ Interview Master Answer

“I would evaluate an LLM across multiple dimensions: knowledge and reasoning, coding, mathematics, long-context understanding, multimodal capabilities, agentic tool use, safety and alignment, and instruction following.”

■ AI Engineer Priority

Focus most on: RAG → Long Context → Agentic/Tool Use → Safety → Instruction Following → Coding.